

Khushi Agarwal

8193820321 | khushiagarwal7821@gmail.com | LinkedIn | GitHub

Full-Stack Software Engineer — Angular/React, Node.js, AWS, Real-Time Systems — 400+ LeetCode problems solved

Education

GLA University, Mathura

June 2021 – 2025

Bachelor of Technology in Computer Science and Engineering — CGPA: 8.16

- Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Operating Systems, Database Management Systems

Experience

Software Engineer Apprentice

June 2025 – June 2026

S&P Global

- Migrated a large-scale Angular application from v8 to v16 across **20+ modules and 100+ components**, resolving breaking changes in routing, RxJS, HttpClient, and third-party libraries
- Upgraded RxJS v6 → v7 and refactored **50+ observable patterns**, implementing `takeUntil` lifecycle management to eliminate subscription-related memory leaks
- Resolved **200+ TypeScript strict-mode errors** and modernized project configuration for Angular 16 compatibility; diagnosed complex UI rendering issues using `NgZone` and `ChangeDetectorRef`
- Designed and maintained a serverless data feed monitoring & alerting system processing **1,000+ daily/intraday feeds** from **2,000+ data sources**
- Automated monitoring workflows using AWS Lambda, Amazon EventBridge, and Amazon SNS, **reducing manual effort by 80%**; enhanced observability using Amazon CloudWatch
- Leveraged AI-assisted development tools to accelerate large-scale refactoring, debugging, and technical documentation

Backend Development Training – SQL & MongoDB (Jovac)

June 2023 – Aug 2023

GLA University

- Completed backend development training focused on SQL and MongoDB, covering database design, queries, data management, and security fundamentals

Projects

Real-Time Collaborative Whiteboard | Angular, Node.js, Express.js, MongoDB, Socket.IO

- Built a full-stack real-time collaborative whiteboard enabling multiple users to draw, edit, and manage shared boards simultaneously via an interactive canvas (pen, eraser, shapes, lines, text)
- Implemented real-time multi-user synchronization using **Socket.IO** and board-specific rooms to broadcast live activity across connected participants with low latency
- Engineered **JWT-based authentication and role-based access control** (owner/collaborator) with a token-based secure invitation and sharing system for onboarding collaborators
- Deployed frontend on Vercel and backend on Render, shipping a production-ready live application — [ai-collaboration-64aq.vercel.app](#)

Algo-Visualizer | React.js, Material-UI, Tailwind CSS

- Built an interactive algorithm visualization platform enabling users to visualize sorting algorithms step-by-step with dynamic animations and interactive games reinforcing time-complexity concepts
- Structured the application into reusable, responsive components using React Router and Vite; deployed on Vercel — [algotvisualizer-beta.vercel.app](#)

RentFashion | Node.js, Express.js, Passport.js, EJS, MongoDB

- Developed a dual-portal rental platform with separate buyer/seller workflows for product management, rental, purchase, cart, and commenting
- Implemented secure authentication and authorization using Passport.js; integrated MongoDB for persistent storage with EJS-based dynamic rendering

Skills

Languages: Java, Python, JavaScript, TypeScript

Frontend: Angular (v8–v16), React.js, RxJS, HTML5, CSS3, Bootstrap, Material-UI

Backend & Real-Time: Node.js, Express.js, RESTful APIs, Socket.IO, WebSockets, JWT, Mongoose

Databases & Cloud: SQL, MongoDB, AWS Lambda, EventBridge, SNS, CloudWatch, Secrets Manager, Vercel, Render

Tools & Practices: Git, GitHub, Postman, VS Code, System Design Fundamentals, Debugging, Performance Optimization

Achievements

- Solved **400+ Data Structures and Algorithms problems** on LeetCode, demonstrating strong problem-solving and algorithmic skills
- Earned a **Gold Medal Certification** for excellence in Backend Development training focused on MongoDB and SQL